



Project

Preface

Notes on planning:

The energy consumption quantities do not take into account light scenes and their dimming levels.

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Site 1

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Room 1

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Site 1 - Building 1 - Storey 1

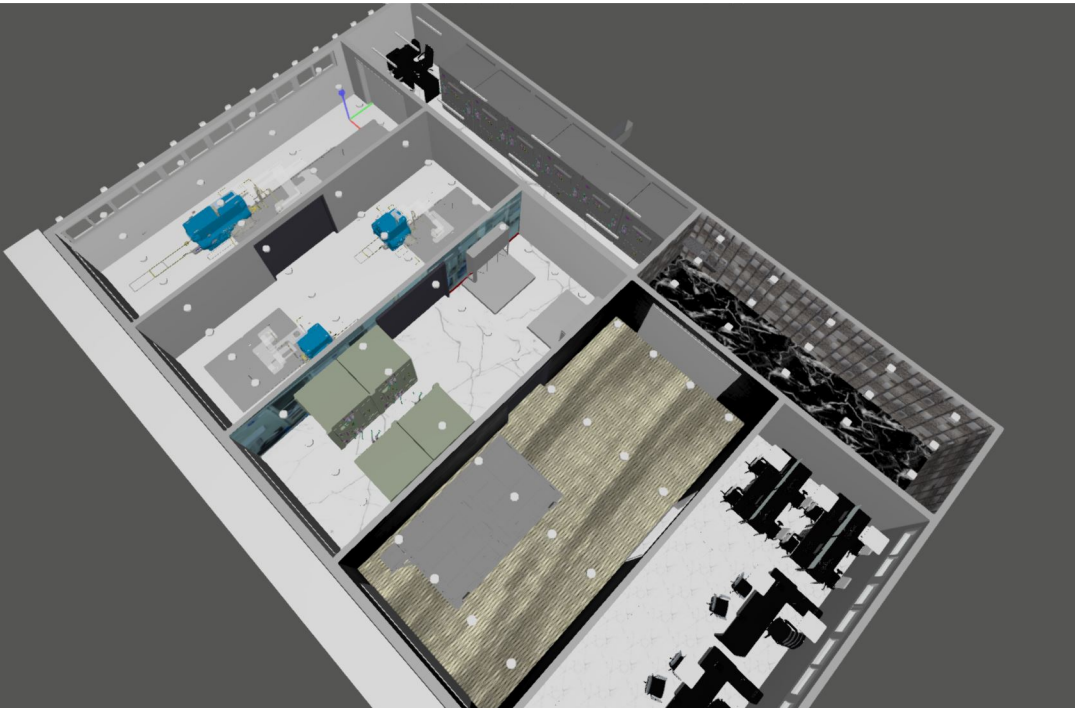
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Glossary55



Description

Luminaire list

Φ_{total} 499618 lm	P_{total} 4751.4 W	Luminous efficacy 105.2 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
12	3F Filippi S.p.A.	22790+A0 1485	3FLP6060UGR-830 (850mA) + ON-OFF	34.0 W	3830 lm	112.6 lm/W
15	3F Filippi S.p.A.	36576	Lucequadro LED 2000 EP VS	21.0 W	2291 lm	109.1 lm/W
64	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W
45	ASTZ	-	DO15-40-301 Kosmos SP 850	39.0 W	4917 lm	126.1 lm/W
9	LIGMAN	NYB- 80636-O- 40	NYBRO 6 Surface ceiling luminaires	92.6 W	2801 lm	30.3 lm/W

Product data sheet

3F Filippi S.p.A. - 3FLP6060UGR-830 (850mA) + ON-OFF



Article No.	22790+A01485
P	34.0 W
Φ_{Lamp}	3830 lm
$\Phi_{Luminaire}$	3830 lm
η	100.00 %
Luminous efficacy	112.6 lm/W
CCT	3000 K
CRI	80

ILLUMINOTECHNICAL

Luminous efficiency 100% (DLOR 100%, ULOR 0%).

Initial luminous flux of the luminaire 3830 lm.

Direct symmetric distribution.

Installation Interdistance Transv.D = $1.18 \times h_u$ - Long.D = $1.30 \times h_u$.

Average luminance <3000 cd/m² for radial angles >65°.

Tabular UGR (CIE 117 - 4H-8H; S=0.25H; 70/50/20): RUG 18.1 - 18.3.

Beam angle: 86° - 90°.

Luminous efficacy 113 lm/W.

Lifetime (L93/B10): 30000 h. (tq+25°C)

Lifetime (L90/B10): 50000 h. (tq+25°C)

Lifetime (L80/B10): 80000 h. (tq+25°C)

Lifetime (L75/B10): 100000 h. (tq+25°C)

Sudden decreased luminous flux after 50000 hours: 0% (C0).

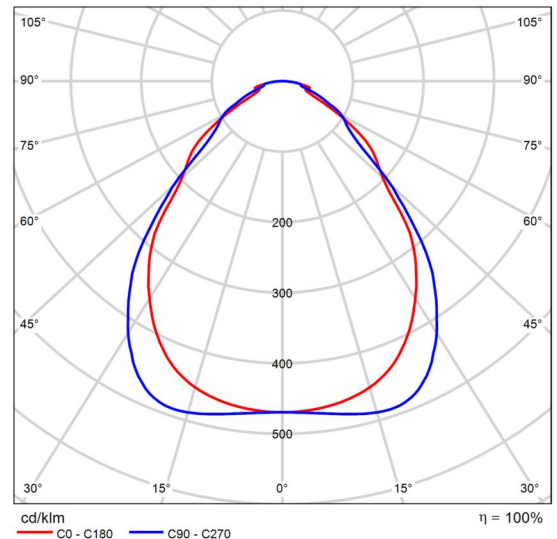
Photobiological safety in compliance with IEC/TR 62778: RG0 risk exempt, (IEC 62471).

In compliance with IEC/EN 62722-2-1 - IEC/EN 62717 standards.

SOURCE

2 linear LED modules 830.

Energy efficiency class (UE 2019/2020 - UE 2019/2015): C.



Polar LDC

Glare evaluation according to UGR												
p Ceiling		70	70	50	50	30	70	70	50	50	30	
p Walls		50	30	50	30	30	50	30	50	30	30	
p Floor		20	20	20	20	20	20	20	20	20	20	
Room size X Y		Viewing direction at right angles to lamp axis					Viewing direction parallel to lamp axis					
2H	2H	15.1	16.2	15.4	16.5	16.7	15.8	17.0	16.1	17.2	17.5	
	3H	15.8	16.8	16.1	17.1	17.4	16.6	17.6	16.9	17.9	18.2	
	4H	16.3	17.3	16.6	17.6	17.9	16.9	17.9	17.2	18.2	18.5	
	6H	16.9	17.8	17.2	18.1	18.4	17.2	18.2	17.6	18.5	18.8	
	8H	17.1	18.0	17.4	18.3	18.6	17.4	18.3	17.8	18.6	18.9	
	12H	17.2	18.1	17.6	18.4	18.8	17.5	18.4	17.9	18.7	19.0	
4H	2H	15.5	16.5	15.8	16.8	17.1	16.1	17.1	16.4	17.3	17.6	
	3H	16.4	17.2	16.8	17.5	17.9	16.9	17.8	17.3	18.1	18.4	
	4H	17.0	17.8	17.4	18.1	18.5	17.5	18.2	17.9	18.6	18.9	
	6H	17.7	18.4	18.2	18.8	19.2	18.0	18.7	18.4	19.0	19.4	
	8H	18.0	18.6	18.4	19.0	19.4	18.2	18.9	18.7	19.2	19.7	
	12H	18.3	18.8	18.7	19.2	19.7	18.4	19.0	18.9	19.4	19.8	
8H	4H	17.2	17.8	17.7	18.2	18.6	17.7	18.3	18.1	18.7	19.1	
	6H	18.1	18.6	18.6	19.0	19.5	18.4	18.9	18.9	19.4	19.8	
	8H	18.5	18.9	19.0	19.4	19.9	18.8	19.2	19.2	19.7	20.1	
	12H	18.8	19.2	19.3	19.7	20.2	19.0	19.4	19.5	19.9	20.4	
12H	4H	17.3	17.8	17.7	18.2	18.7	17.7	18.3	18.1	18.7	19.1	
	6H	18.2	18.6	18.7	19.1	19.6	18.5	18.9	19.0	19.4	19.9	
	8H	18.6	19.0	19.1	19.5	20.0	18.9	19.3	19.4	19.7	20.2	
Variation of the observer position for the luminaire distances S												
S = 1.0H		+0.4 / -0.3					+0.4 / -0.5					
S = 1.5H		+0.4 / -0.9					+0.6 / -0.9					
S = 2.0H		+0.8 / -1.6					+1.3 / -1.4					
Standard table		BK05					BK04					
Correction summand		0.9					0.9					
Corrected glare indices referring to 3830lm Total luminous flux												

UGR diagram (SHR: 0.25)

Product data sheet

3F Filippi S.p.A. - 3FLP6060UGR-830 (850mA) + ON-OFF

CIE 13.3 Colour rendering index: CRI >80 (R9 <50%).

IES TM-30 Fidelity Index: Rf = 82 Rg = 95.

CCT nominal colour temperature 3000 K.

Colour initial tolerance (MacAdam): SDCM 3.

MECHANICAL

Housing in white painted epoxy-polyester powder aluminium.

Diffuser in SMP transparent microprismatic methacrylate (PMMA) externally, anti-glare with high transmittance.

Perimetral frame in white polycarbonate.

Anti-fall safety cable.

Luminaire with limited surface temperature. - D - (EN 60598-2-24)

Dimensions: 595x595 mm, height 9 mm. Weight 2.505 kg.

IP43 protection degree for exposed part, IP20 for recessed part.

Mechanical strength to impacts IK06 (1 joule).

Glow-wire test resistance 650°C.

ELECTRICAL

Wiring on a separate unit (Multi-current power supply, to be ordered separately).

Power of the luminaire 34 W.

CE - IEC 60598-1 - EN 60598-1.

SAFE FLICKER: PstLM=<1 and SVM=<0.4 (IEC TR 61547-1 and IEC TR 63158), to ensure a more comfortable and safe light.

Luminaire compliant with EN 60598-2-22 for power supply from a centralised emergency system CPSS (Central Power Supply System), not incorporated in the luminaire - high risk areas excluded. The default power and flux are 100% in AC and 100% in DC.

Ambient temperature from 0°C to +25°C.

Temperature class T6 max 85°C.

Relative humidity UR: <85%.

INSTALLATION

Lay-in recessed fitting / Pull-up recessed fitting with brackets / Ceiling installation with frame / Suspension via accessory.

All accessories dedicated to this product are available on the Catalog and on our website www.3F-Filippi.com.

ACCESSORIES

A01485 - ZK700-900EL DRIVER ON-OFF DIP-SWITCH.

Electronic wiring 230V-50/60Hz, power factor 0.95, constant output current, SELV, class II, 1 driver.

Multi-current power supply that allows you to choose the driving current of the fixture at the time of installation according to the required illuminance.

Quick connection non-reversible plug-socket terminal block, also for cascade connection.

APPLICATIONS

Product data sheet

3F Filippi S.p.A. - 3FLP6060UGR-830 (850mA) + ON-OFF

Suitable product for food production plants (HACCP), IFS (Food Version 6), BRC (GSFS Food Version 7).

Environments: with VDTs, meeting rooms, offices.

Environments: recreational, transit areas, corridors, schools, stairwells.

Environments where soft diffuse light is required for optimal visual comfort.

WARNING

Luminaire designed for disposal/recycling at end-of-life.

Replaceable (LED only) light source by a professional. Replaceable control gear by a professional.

Product data sheet

3F Filippi S.p.A. - Lucequadro LED 2000 EP VS



Article No.	36576
P	21.0 W
Φ_{Lamp}	2291 lm
$\Phi_{Luminaire}$	2291 lm
η	100.00 %
Luminous efficacy	109.1 lm/W
CCT	3991 K
CRI	80

ILLUMINOTECHNICAL

Luminous efficiency 100% (DLOR 100%, ULOR 0%).

Initial luminous flux of the luminaire 2291 lm.

Emergency luminaire luminous flux BLF 20%.

Direct symmetric distribution.

Installation Interdistance Transv.D = $0.83 \times h_u$ - Long.D = $0.80 \times h_u$.

Average luminance <3000 cd/m² for radial angles >65°.

Tabular UGR (CIE 117 - 4H-8H; S=0.25H; 70/50/20): RUG 17.2 - 17.

Beam angle: 58° - 57°.

Luminous efficacy 109 lm/W.

Lifetime (L90/B10): 30000 h. (tq+25°C)

Lifetime (L85/B10): 50000 h. (tq+25°C)

Lifetime (L70/B10): 80000 h. (tq+25°C)

Sudden decreased luminous flux after 50000 hours: 0% (C0).

Photobiological safety in compliance with IEC/TR 62778: RG1 low risk, (IEC 62471).

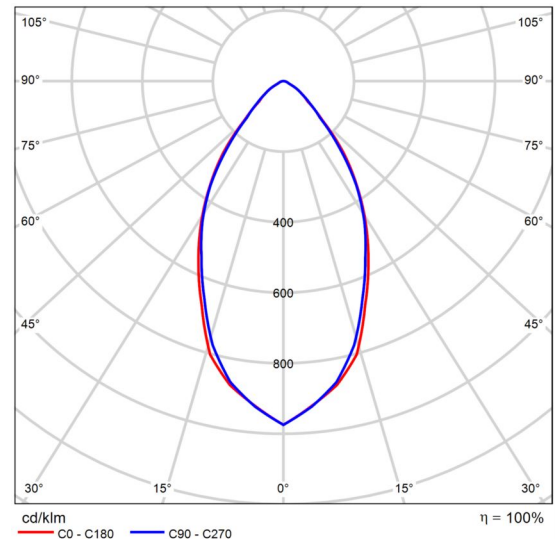
In compliance with IEC/EN 62722-2-1 - IEC/EN 62717 standards.

SOURCE

Compact LED module 2000/840.

Energy efficiency class (UE 2019/2020 - UE 2019/2015): D.

CIE 13.3 Colour rendering index: CRI >80 (R9 <50%).



Polar LDC

Glare evaluation according to UGR												
p Ceiling		70	70	50	50	30	70	70	50	50	30	
p Walls		50	30	50	30	30	50	30	50	30	30	
p Floor		20	20	20	20	20	20	20	20	20	20	
Room size X Y		Viewing direction at right angles to lamp axis					Viewing direction parallel to lamp axis					
2H	2H	16.8	17.8	17.1	18.0	18.2	16.7	17.7	17.0	17.9	18.1	
	3H	16.9	17.8	17.2	18.0	18.3	16.8	17.6	17.1	17.9	18.1	
	4H	16.9	17.7	17.2	18.0	18.3	16.8	17.6	17.1	17.9	18.1	
	6H	16.9	17.7	17.3	18.0	18.3	16.8	17.5	17.1	17.8	18.1	
	8H	17.0	17.7	17.3	18.0	18.3	16.8	17.5	17.1	17.8	18.1	
	12H	16.9	17.7	17.3	18.0	18.3	16.8	17.5	17.1	17.8	18.1	
4H	2H	16.8	17.7	17.2	17.9	18.2	16.7	17.6	17.1	17.8	18.1	
	3H	17.0	17.6	17.3	18.0	18.3	16.8	17.5	17.2	17.8	18.2	
	4H	17.0	17.7	17.4	18.0	18.4	16.9	17.5	17.3	17.9	18.2	
	6H	17.1	17.7	17.5	18.0	18.4	17.0	17.5	17.4	17.9	18.3	
	8H	17.2	17.7	17.6	18.0	18.5	17.0	17.5	17.4	17.9	18.3	
	12H	17.2	17.6	17.6	18.0	18.5	17.0	17.4	17.4	17.8	18.3	
8H	4H	17.0	17.5	17.4	17.9	18.3	16.9	17.4	17.3	17.7	18.2	
	6H	17.1	17.6	17.6	18.0	18.4	17.0	17.4	17.4	17.8	18.2	
	8H	17.2	17.6	17.7	18.0	18.5	17.0	17.4	17.5	17.8	18.3	
	12H	17.3	17.6	17.8	18.0	18.5	17.1	17.4	17.5	17.8	18.3	
	4H	17.0	17.4	17.4	17.8	18.3	16.8	17.3	17.3	17.7	18.1	
	6H	17.1	17.5	17.6	17.9	18.4	17.0	17.3	17.4	17.8	18.2	
12H	8H	17.2	17.5	17.7	18.0	18.5	17.0	17.3	17.5	17.8	18.3	
	12H	17.2	17.5	17.7	18.0	18.5	17.0	17.3	17.5	17.8	18.3	
Variation of the observer position for the luminaire distances S												
S = 1.0H		+1.3 / -1.9					+1.4 / -1.9					
S = 1.5H		+2.9 / -3.4					+2.9 / -3.4					
S = 2.0H		+4.6 / -4.6					+4.7 / -4.9					
Standard table		BK01					BK01					
Correction summand		-0.9					-1.0					
Corrected glare indices referring to 2291lm Total luminous flux												

UGR diagram (SHR: 0.25)

Product data sheet

3F Filippi S.p.A. - Lucequadro LED 2000 EP VS

IES TM-30 Fidelity Index: $R_f = 84$ $R_g = 95$.
CCT nominal colour temperature 4000 K.
Colour initial tolerance (MacAdam): SDCM 3.
Zhaga Book 3 compliant.

MECHANICAL

Housing in white painted steel.
Passive heat dissipator in anodised aluminium, oversized, for optimum thermal management of the LED module.
Flow recuperator in specular aluminium with superficial titanium-magnesium treatment, non-iridescent.
Moulded tempered anti-glare non-combustible glass, thickness 4 mm.
Lock-in mounting of the glass in the lateral seats in specular aluminium.
Fixing brackets in galvanised steel.
Slot-cover polycarbonate casings.
Dimensions: 235x235 mm, height 116 mm. Weight 3.15 kg.
IP44 protection degree for exposed part, IP20 for recessed part.
Mechanical strength to impacts IK08 (5 joule).
Glow-wire test resistance 960°C.

ELECTRICAL

Wiring on a separate unit.
Halogen Free electronic wiring 230V-50/60Hz, power factor 0.90, THD <25%, constant output current, SELV, class II, 1 driver.
Power of the luminaire 21 W.
CE - IEC 60598-1 - EN 60598-1.
EP maintained emergency wiring on board, 1h duration, 24h recharge; compliant with EN 60598-2-22, excluding high risk areas.
SAFE FLICKER: $P_{stLM} < 1$ and $SVM < 0.4$ (IEC TR 61547-1 and IEC TR 63158), to ensure a more comfortable and safe light.
Ambient temperature from +5°C to +25°C.
Temperature class T6 max 85°C.
Relative humidity UR: <85%.

INSTALLATION

Pull-up recessed fitting.
False ceiling carving: 210x210 mm.
All accessories dedicated to this product are available on the Catalog and on our website www.3F-Filippi.com.

ACCESSORIES

A0189 - Pair of reinforcing brackets for mineral fibre, metal and plasterboard panels, height 20 mm.

APPLICATIONS

Transit areas, corridors, commercial environments, display areas, shops and shop windows.

Product data sheet

3F Filippi S.p.A. - Lucequadro LED 2000 EP VS

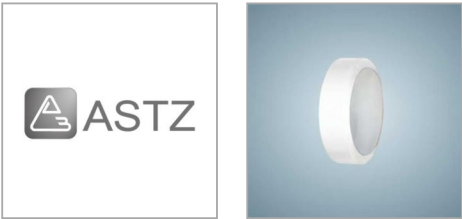
WARNING

Luminaire designed for disposal/recycling at end-of-life.

Replaceable (LED only) light source by a professional. Replaceable control gear by a professional.

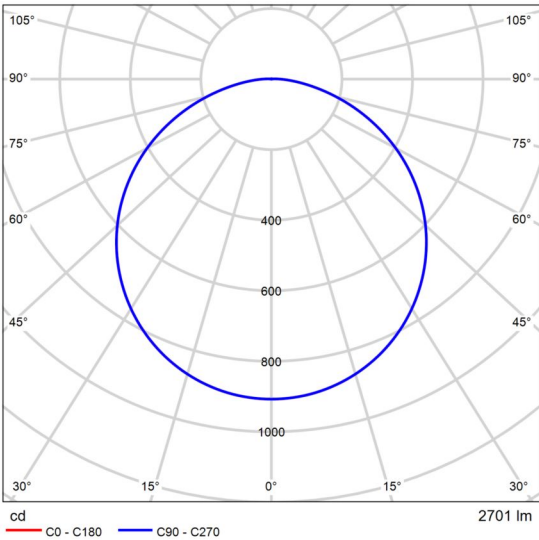
Product data sheet

ASTZ - DBO85-24-241 Tablette EM1 840



Article No.	-
P	22.5 W
Φ _{Lamp}	-
Φ _{Luminaire}	2701 lm
η	-
Luminous efficacy	120.2 lm/W
CCT	4000 K
CRI	80

ДБО85-24-241 Tablette EM1 840
ТУ 3461-043-05014337-2009
Предназначены для освещения общественных и
производственных помещений, торговых залов, гипермаркетов,
складских помещений, муниципальных объектов ЖКХ.
220 В, 50 Гц, 24Вт LED модуль, 4000 (3700-4260)К, Ra 80, драйвер
(PF=0,98), IP65, У1
Световой поток светильника: 2701 лм
Класс защиты от поражения электрическим током: I DхН=266х82
мм. Масса: 1,92 кг
Корпус из алюминия.
Рассеиватель из матового поликарбоната.
Установлен блок аварийного питания. Время работы в
аварийном режиме до 1 часа. Световой поток в аварийном
режиме - 10%.
Установка: Крепление на опорную поверхность.



Polar LDC

Glare evaluation according to UGR												
		70	70	50	50	30	70	70	50	50	30	30
p Ceiling		50	30	50	30	30	50	30	50	30	30	30
p Walls		50	30	50	30	30	50	30	50	30	30	30
p Floor		20	20	20	20	20	20	20	20	20	20	20
Room size X Y		Viewing direction at right angles to lamp axis					Viewing direction parallel to lamp axis					
2H	2H	20.6	22.0	20.9	22.2	22.5	20.6	22.0	20.9	22.2	22.5	
	3H	21.9	23.2	22.3	23.4	23.7	21.9	23.2	22.3	23.4	23.7	
	4H	22.4	23.6	22.7	23.9	24.2	22.4	23.6	22.7	23.9	24.2	
	6H	22.7	23.8	23.0	24.1	24.4	22.7	23.8	23.0	24.1	24.4	
	8H	22.7	23.8	23.1	24.1	24.5	22.7	23.8	23.1	24.1	24.5	
4H	12H	22.8	23.8	23.2	24.1	24.5	22.8	23.8	23.2	24.1	24.5	
	2H	21.2	22.4	21.6	22.7	23.0	21.2	22.4	21.6	22.7	23.0	
	3H	22.7	23.7	23.1	24.0	24.4	22.7	23.7	23.1	24.0	24.4	
	4H	23.3	24.2	23.7	24.5	24.9	23.3	24.2	23.7	24.5	24.9	
	6H	23.6	24.4	24.1	24.8	25.3	23.6	24.4	24.1	24.8	25.3	
8H	12H	23.8	24.5	24.2	24.9	25.3	23.8	24.5	24.2	24.9	25.3	
	2H	23.8	24.5	24.3	24.9	25.4	23.8	24.5	24.3	24.9	25.4	
	4H	23.5	24.2	23.9	24.6	25.1	23.5	24.2	23.9	24.6	25.1	
	6H	24.0	24.6	24.4	25.0	25.5	24.0	24.6	24.4	25.0	25.5	
	8H	24.1	24.7	24.6	25.2	25.6	24.1	24.7	24.6	25.2	25.6	
12H	12H	24.3	24.7	24.8	25.2	25.7	24.3	24.7	24.8	25.2	25.7	
	4H	23.5	24.2	23.9	24.6	25.0	23.5	24.2	23.9	24.6	25.0	
	6H	24.0	24.5	24.5	25.0	25.5	24.0	24.5	24.5	25.0	25.5	
8H	24.2	24.7	24.7	25.2	25.7	24.2	24.7	24.7	25.2	25.7		
Variation of the observer position for the luminaire distances S												
S = 1.0H		+0.1 / -0.2					+0.1 / -0.2					
S = 1.5H		+0.3 / -0.5					+0.3 / -0.5					
S = 2.0H		+0.5 / -0.9					+0.5 / -0.9					
Standard table		BK05					BK05					
Correction summand		6.9					6.9					
Corrected glare indices referring to 2701lm Total luminous flux												

UGR diagram (SHR: 0.25)

Product data sheet

ASTZ - DO15-40-301 Kosmos SP 850



Article No.	-
P	39.0 W
Φ_{Lamp}	-
$\Phi_{\text{Luminaire}}$	4917 lm
η	-
Luminous efficacy	126.1 lm/W
CCT	5000 K
CRI	80

ДО15-40-301 Kosmos SP 850

ТУ 3461-050-05014337-2014

Предназначены для освещения площадей, стадионов, фасадов зданий, архитектурных памятников, подъездных путей, строительных площадок и других открытых пространств, а также внутреннего освещения закрытых спортивных и других сооружений.

220 В, 50 Гц, LED модуль, 39 Вт, 5000К, Ra=80, драйвер (PF=0,96), IP65, У1

Световой поток светильника: 4916 лм

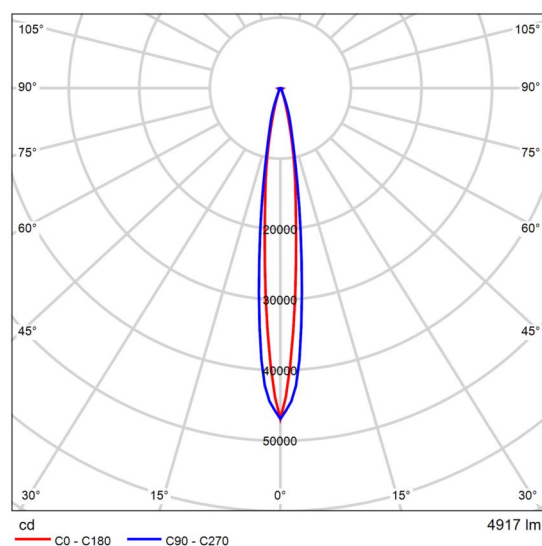
Осевая сила света: 42295 кд

КСС "К" (12°)

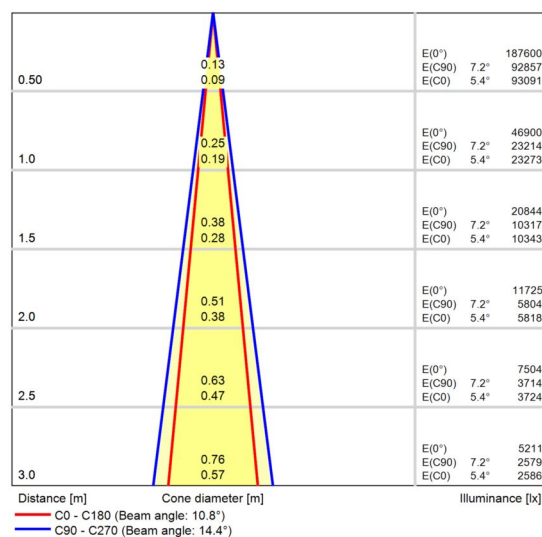
Класс защиты от поражения электрическим током: I LxH=230x120 мм. Масса: 3,2 кг

Корпус радиатор из анодированного алюминия.

Установка: Светильник монтируется на опорную поверхность из несгораемого материала.



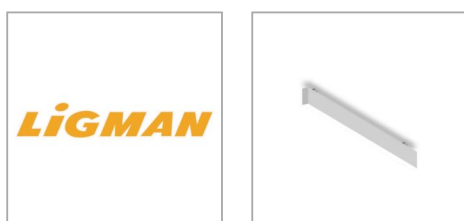
Polar LDC



Cone diagram

Product data sheet

LIGMAN - NYBRO 6 Surface ceiling luminaires



Article No.	NYB-80636-O-40
P	92.6 W
Φ_{Lamp}	–
$\Phi_{\text{Luminaire}}$	2801 lm
η	–
Luminous efficacy	30.3 lm/W
CCT	– 7247 K 4032 K
CRI	– -18 80

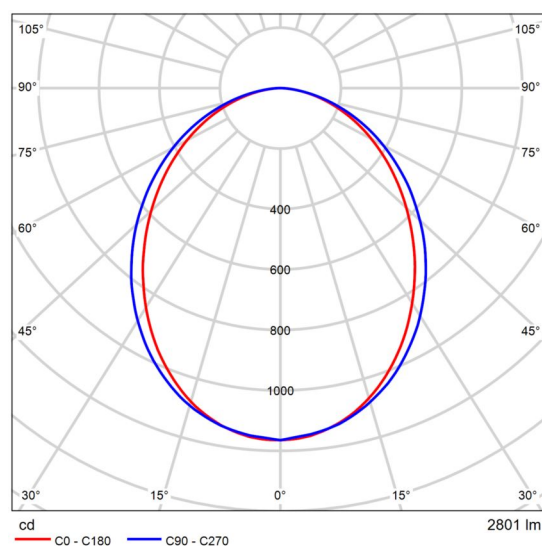
- Lamp LED
- CRI Ra > 80
- MacAdam Ellipse 3 SDCM
- Colour Temperature RGBW40
- Diffuser for light output control O – Opal diffuser UGR <19
P – Micro-prismatic UGR <13

Surface, recessed or pendant mounted wall and ceiling linear luminaires with symmetrical up and down lighters for general or specialized lighting application

Lighting control system.

- Wireless control employing the Bluetooth® 4.0 protocol controlled via Ligman's mobile App or physical control via xPress switch
- Daylight and occupancy sensor options

Low copper content die casted aluminium end caps with



Polar LDC

Product data sheet

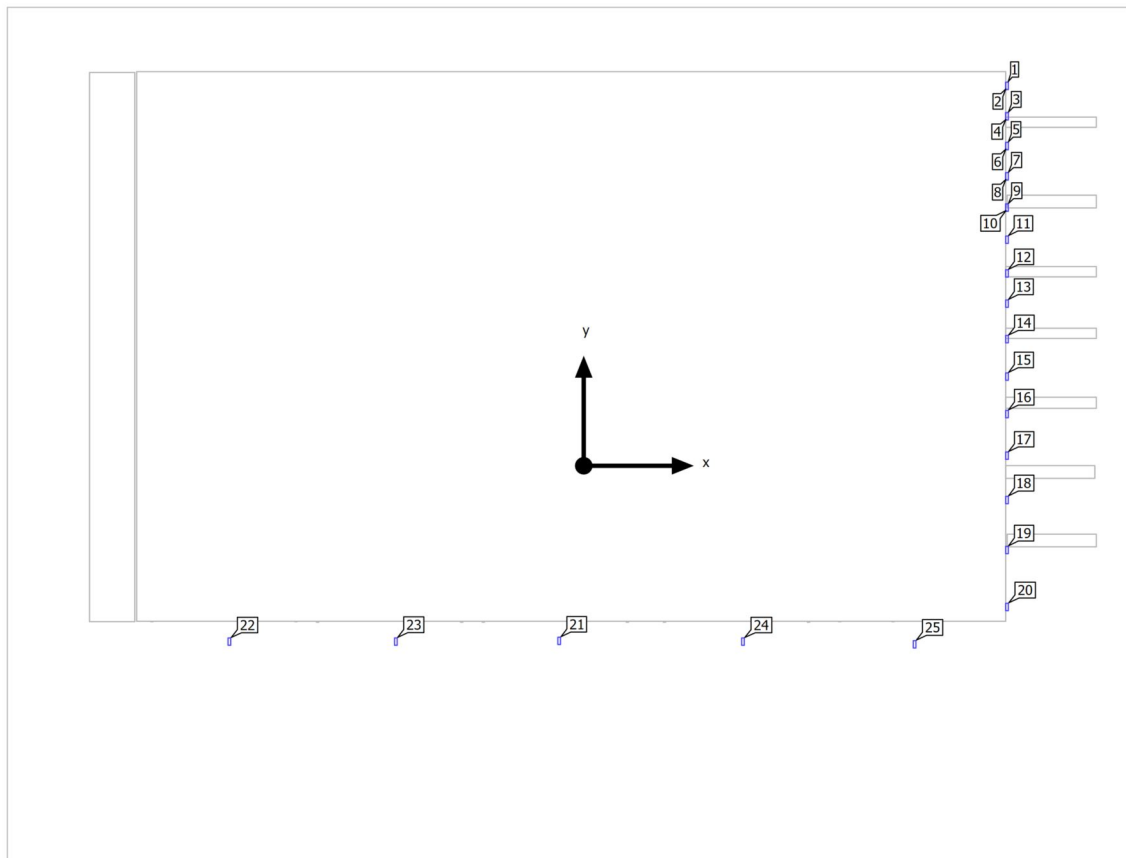
LIGMAN - NYBRO 6 Surface ceiling luminaires

aluminium profiles completed with 316 grade stainless steel fasteners. Option for opal or micro-prismatic PMMA diffusers for glare control.

Built-in driver. Passive thermal management. Surface wall and ceiling mountings with the use of steel support and screws. Pendant mounting with the application of adjustable steel suspension (1m) complete with electrical wires in white.

Applications: private residences, offices schools and universities, retail, showrooms, galleries, museums, conference halls, meeting rooms, banks, bars, restaurants, hotel and similar installations

Site 1

Luminaire layout plan

Site 1

Luminaire layout plan



Manufacturer	ASTZ	P	39.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	4917 lm
Article name	DO15-40-301 Kosmos SP 850		
Fitting	1x DO15-40-301 Kosmos SP 850		

Individual luminaires

X	Y	Mounting height	Luminaire
13.292 m	11.934 m	2.951 m	1
13.292 m	11.934 m	2.951 m	2
13.292 m	10.978 m	2.951 m	3
13.292 m	10.978 m	2.951 m	4
13.292 m	10.043 m	2.951 m	5
13.292 m	10.043 m	2.951 m	6
13.292 m	9.091 m	2.951 m	7
13.292 m	9.091 m	2.951 m	8
13.292 m	8.111 m	2.951 m	9
13.292 m	8.111 m	2.951 m	10
13.292 m	7.101 m	2.951 m	11
13.292 m	6.042 m	2.951 m	12
13.292 m	5.093 m	2.951 m	13

Site 1

Luminaire layout plan

X	Y	Mounting height	Luminaire
13.292 m	3.978 m	2.951 m	14
13.292 m	2.801 m	2.951 m	15
13.292 m	1.625 m	2.951 m	16
13.292 m	0.317 m	2.951 m	17
13.292 m	-1.076 m	2.951 m	18
13.292 m	-2.645 m	2.951 m	19
13.292 m	-4.435 m	2.951 m	20
-0.771 m	-5.500 m	2.800 m	21
-11.128 m	-5.523 m	2.800 m	22
-5.900 m	-5.523 m	2.800 m	23
5.000 m	-5.523 m	2.800 m	24
10.390 m	-5.608 m	2.800 m	25

Site 1

Luminaire list

Φ_{total} 122900 lm	P_{total} 975.0 W	Luminous efficacy 126.1 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
25	ASTZ	-	DO15-40-301 Kosmos SP 850	39.0 W	4917 lm	126.1 lm/W

Building 1

Luminaire list Φ_{total}

376718 lm

 P_{total}

3776.4 W

Luminous efficacy

99.8 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
12	3F Filippi S.p.A.	22790+A0 1485	3FLP6060UGR-830 (850mA) + ON-OFF	34.0 W	3830 lm	112.6 lm/W
15	3F Filippi S.p.A.	36576	Lucequadro LED 2000 EP VS	21.0 W	2291 lm	109.1 lm/W
64	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W
20	ASTZ	-	DO15-40-301 Kosmos SP 850	39.0 W	4917 lm	126.1 lm/W
9	LIGMAN	NYB-80636-O-40	NYBRO 6 Surface ceiling luminaires	92.6 W	2801 lm	30.3 lm/W

Building 1 · Storey 1 (Light scene 1)

Room list

Room 5			Room 6	
Room 1	Room 2	Room 4	Room 7	Room 8

Building 1 · Storey 1 (Light scene 1)

Room list

Room 1

P_{total}
360.0 W

A_{Room}
65.12 m²

Lighting power density
5.53 W/m² (Space)
7.48 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm

Room 2

P_{total}
360.0 W

A_{Room}
65.00 m²

Lighting power density
5.54 W/m² (Space)
7.50 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm

Room 4

P_{total}
360.0 W

A_{Room}
64.63 m²

Lighting power density
5.57 W/m² (Space)
7.55 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm

Building 1 · Storey 1 (Light scene 1)

Room list

Room 5

P_{total}
833.4 W**A_{Room}**
54.67 m²**Lighting power density**
15.24 W/m² (Space)
22.70 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
9	LIGMAN	NYB-80636-O-40	NYBRO 6 Surface ceiling luminaires	92.6 W	2801 lm

Room 6

P_{total}
315.0 W**A_{Room}**
41.37 m²**Lighting power density**
7.61 W/m² (Space)
11.48 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
15	3F Filippi S.p.A.	36576	Lucequadro LED 2000 EP VS	21.0 W	2291 lm

Room 7

P_{total}
360.0 W**A_{Room}**
77.40 m²**Lighting power density**
4.65 W/m² (Space)
6.05 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm

Building 1 · Storey 1 (Light scene 1)

Room list

Room 8

 P_{total}
408.0 W A_{Room}
64.89 m²**Lighting power density**
6.29 W/m² (Space)
8.50 W/m² (Working plane)

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
12	3F Filippi S.p.A.	22790+A0 1485	3FLP6060UGR-830 (850mA) + ON-OFF	34.0 W	3830 lm

Building 1 · Storey 1

Luminaire list Φ_{total}

376718 lm

 P_{total}

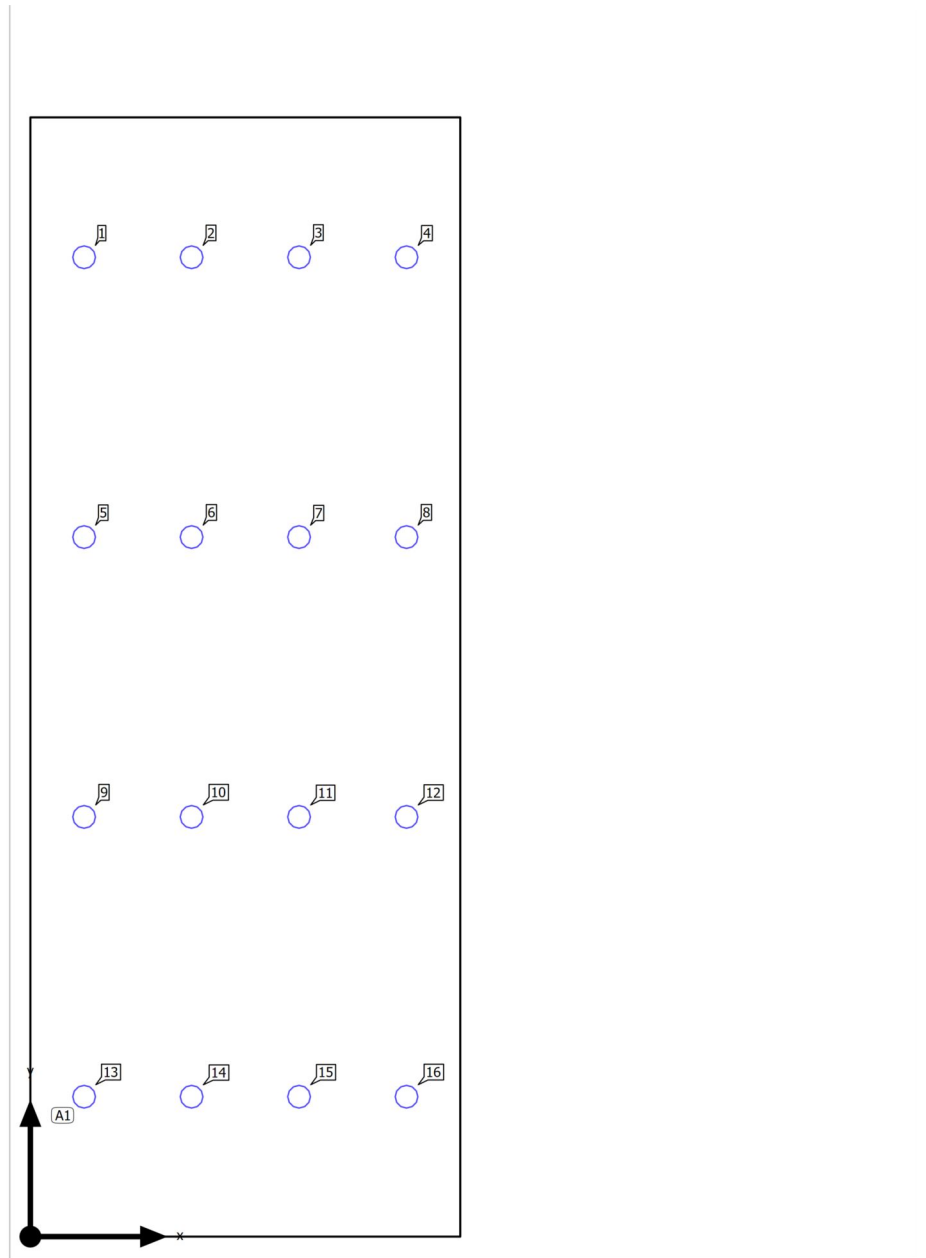
3776.4 W

Luminous efficacy

99.8 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
12	3F Filippi S.p.A.	22790+A01485	3FLP6060UGR-830 (850mA) + ON-OFF	34.0 W	3830 lm	112.6 lm/W
15	3F Filippi S.p.A.	36576	Lucequadro LED 2000 EP VS	21.0 W	2291 lm	109.1 lm/W
64	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W
20	ASTZ	-	DO15-40-301 Kosmos SP 850	39.0 W	4917 lm	126.1 lm/W
9	LIGMAN	NYB-80636-O-40	NYBRO 6 Surface ceiling luminaires	92.6 W	2801 lm	30.3 lm/W

Building 1 · Storey 1 · Room 1

Luminaire layout plan

Building 1 · Storey 1 · Room 1

Luminaire layout plan

Manufacturer	ASTZ	P	22.5 W
Article No.	-	$\Phi_{\text{Luminaire}}$	2701 lm
Article name	DBO85-24-241 Tablette EM1 840		
Fitting	1x LED		

16 x ASTZ DBO85-24-241 Tablette EM1 840

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.625 m / 1.627 m / 2.800 m	0.625 m	11.392 m	2.800 m	1
X-direction	4 pcs., Centre - centre, 1.250 m	1.876 m	11.392 m	2.800 m	2
		3.126 m	11.392 m	2.800 m	3
Y-direction	4 pcs., Centre - centre, 3.255 m	4.376 m	11.392 m	2.800 m	4
		0.625 m	8.137 m	2.800 m	5
Arrangement	A1	1.876 m	8.137 m	2.800 m	6
		3.126 m	8.137 m	2.800 m	7
		4.376 m	8.137 m	2.800 m	8
		0.625 m	4.882 m	2.800 m	9
		1.876 m	4.882 m	2.800 m	10
		3.126 m	4.882 m	2.800 m	11
		4.376 m	4.882 m	2.800 m	12
		0.625 m	1.627 m	2.800 m	13

Building 1 · Storey 1 · Room 1

Luminaire layout plan

X	Y	Mounting height	Luminaire
1.876 m	1.627 m	2.800 m	14
3.126 m	1.627 m	2.800 m	15
4.376 m	1.627 m	2.800 m	16

Building 1 · Storey 1 · Room 1

Luminaire list Φ_{total}

43216 lm

 P_{total}

360.0 W

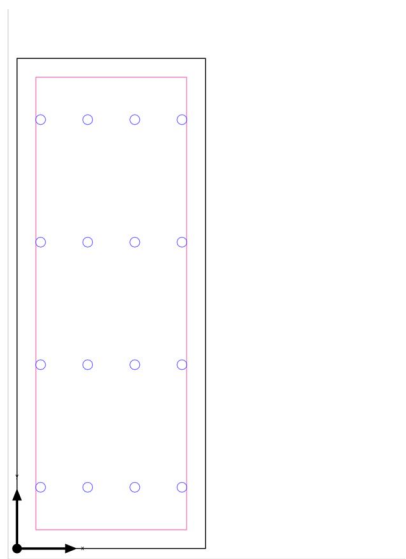
Luminous efficacy

120.0 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W

Building 1 · Storey 1 · Room 1

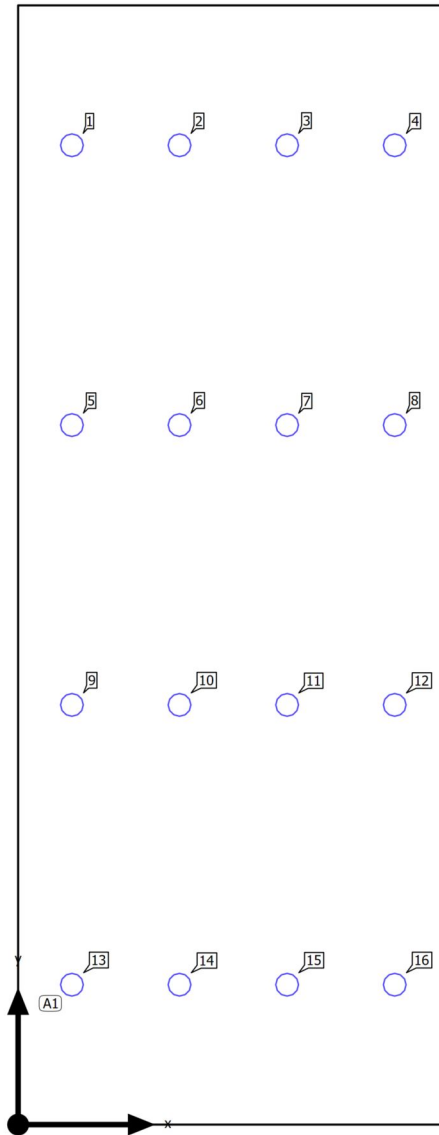
Energy evaluation year



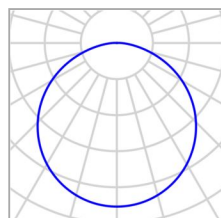
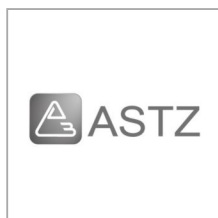
Planning data

Utilisation profile	DIALux presetting (34.2 Standard (office))
Target illuminance	500 lx
Utilisation times	8:00 - 16:00 o'clock
Days per Week	6 (Mon - Sat)

Building 1 · Storey 1 · Room 2

Luminaire layout plan

Building 1 · Storey 1 · Room 2

Luminaire layout plan

Manufacturer	ASTZ	P	22.5 W
Article No.	-	$\Phi_{\text{Luminaire}}$	2701 lm
Article name	DBO85-24-241 Tablette EM1 840		
Fitting	1x LED		

16 x ASTZ DBO85-24-241 Tablette EM1 840

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.625 m / 1.625 m / 2.800 m	0.625 m	11.375 m	2.800 m	1
X-direction	4 pcs., Centre - centre, 1.250 m	1.875 m	11.375 m	2.800 m	2
		3.125 m	11.375 m	2.800 m	3
Y-direction	4 pcs., Centre - centre, 3.250 m	4.375 m	11.375 m	2.800 m	4
		0.625 m	8.125 m	2.800 m	5
Arrangement	A1	1.875 m	8.125 m	2.800 m	6
		3.125 m	8.125 m	2.800 m	7
		4.375 m	8.125 m	2.800 m	8
		0.625 m	4.875 m	2.800 m	9
		1.875 m	4.875 m	2.800 m	10
		3.125 m	4.875 m	2.800 m	11
		4.375 m	4.875 m	2.800 m	12
		0.625 m	1.625 m	2.800 m	13

Building 1 · Storey 1 · Room 2

Luminaire layout plan

X	Y	Mounting height	Luminaire
1.875 m	1.625 m	2.800 m	14
3.125 m	1.625 m	2.800 m	15
4.375 m	1.625 m	2.800 m	16

Building 1 · Storey 1 · Room 2

Luminaire list Φ_{total}

43216 lm

 P_{total}

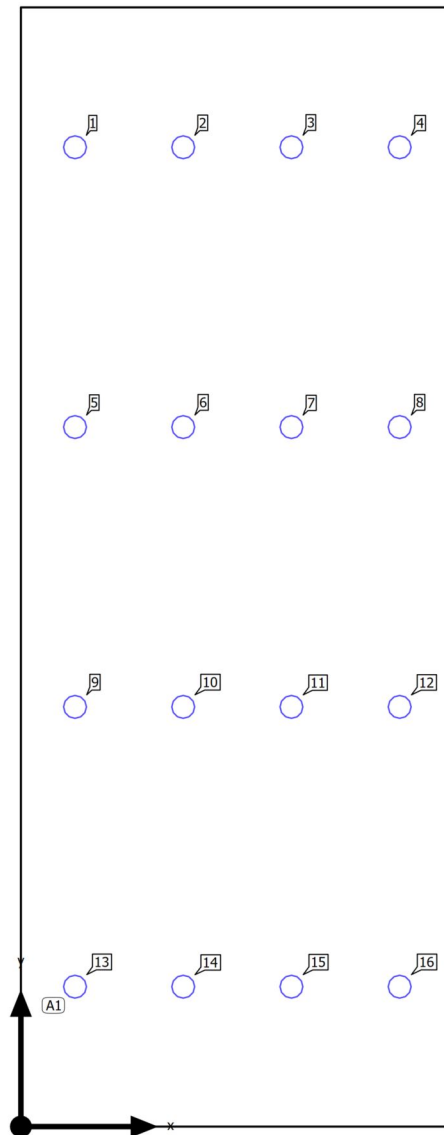
360.0 W

Luminous efficacy

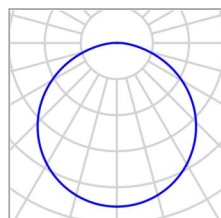
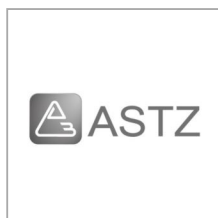
120.0 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W

Building 1 · Storey 1 · Room 4

Luminaire layout plan

Building 1 · Storey 1 · Room 4

Luminaire layout plan

Manufacturer	ASTZ	P	22.5 W
Article No.	-	$\Phi_{\text{Luminaire}}$	2701 lm
Article name	DBO85-24-241 Tablette EM1 840		
Fitting	1x LED		

16 x ASTZ DBO85-24-241 Tablette EM1 840

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.625 m / 1.616 m / 2.800 m	0.625 m	11.310 m	2.800 m	1
X-direction	4 pcs., Centre - centre, 1.250 m	1.875 m	11.310 m	2.800 m	2
		3.125 m	11.310 m	2.800 m	3
Y-direction	4 pcs., Centre - centre, 3.232 m	4.375 m	11.310 m	2.800 m	4
Arrangement	A1	0.625 m	8.079 m	2.800 m	5
		1.875 m	8.079 m	2.800 m	6
		3.125 m	8.079 m	2.800 m	7
		4.375 m	8.079 m	2.800 m	8
		0.625 m	4.847 m	2.800 m	9
		1.875 m	4.847 m	2.800 m	10
		3.125 m	4.847 m	2.800 m	11
		4.375 m	4.847 m	2.800 m	12
		0.625 m	1.616 m	2.800 m	13

Building 1 · Storey 1 · Room 4

Luminaire layout plan

X	Y	Mounting height	Luminaire
1.875 m	1.616 m	2.800 m	14
3.125 m	1.616 m	2.800 m	15
4.375 m	1.616 m	2.800 m	16

Building 1 · Storey 1 · Room 4

Luminaire list Φ_{total}

43216 lm

 P_{total}

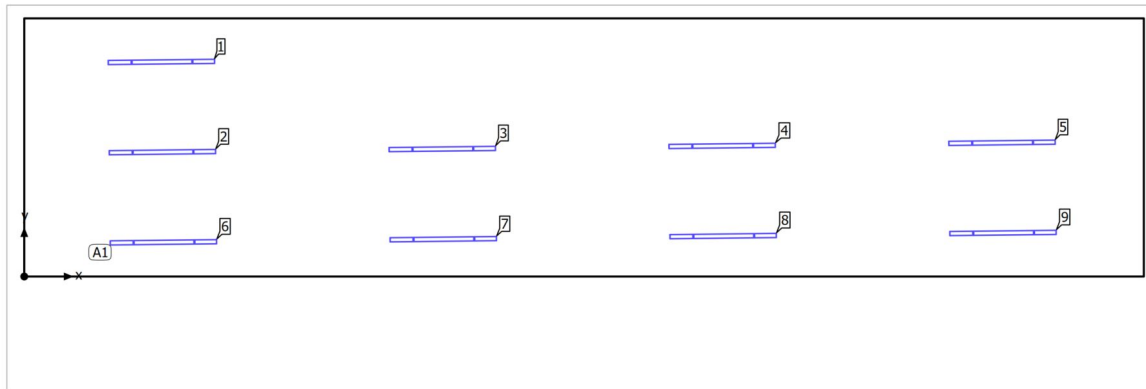
360.0 W

Luminous efficacy

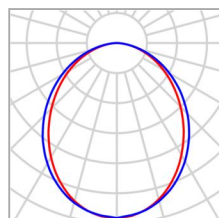
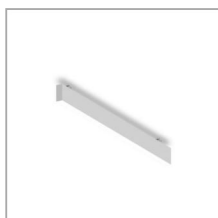
120.0 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W

Building 1 · Storey 1 · Room 5

Luminaire layout plan

Building 1 · Storey 1 · Room 5

Luminaire layout plan

Manufacturer	LIGMAN	P	92.6 W
Article No.	NYB-80636-O-40	$\Phi_{\text{Luminaire}}$	2801 lm
Article name	NYBRO 6 Surface ceiling luminaires		
Fitting	1x NYB-80636-O-R, 1x NYB-80636-O-G, 1x NYB-80636-O-B, 1x NYB-80636-O-W40		

9 x LIGMAN NYBRO 6 Surface ceiling luminaires

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.920 m / 0.470 m / 2.799 m	1.892 m	2.951 m	2.799 m	1
X-direction	4 pcs., Centre - centre, 3.850 m	1.906 m	1.710 m	2.799 m	2
		5.755 m	1.753 m	2.799 m	3
Y-direction	3 pcs., Centre - centre, 1.240 m	9.605 m	1.796 m	2.799 m	4
Arrangement	A1	13.455 m	1.839 m	2.799 m	5
		1.920 m	0.470 m	2.799 m	6
		5.769 m	0.513 m	2.799 m	7
		9.619 m	0.556 m	2.799 m	8
		13.468 m	0.599 m	2.799 m	9

Building 1 · Storey 1 · Room 5

Luminaire list Φ_{total}

25209 lm

 P_{total}

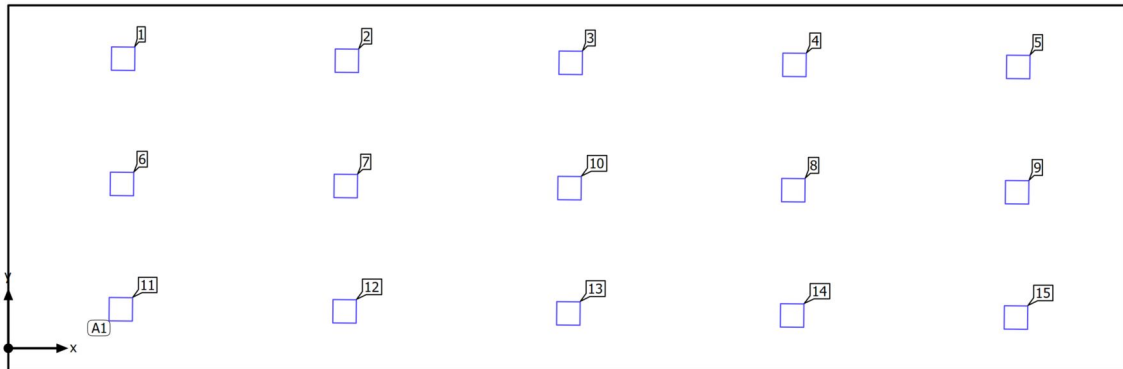
833.4 W

Luminous efficacy

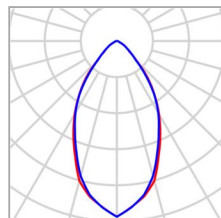
30.2 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
9	LIGMAN	NYB-80636-O-40	NYBRO 6 Surface ceiling luminaires	92.6 W	2801 lm	30.3 lm/W

Building 1 · Storey 1 · Room 6

Luminaire layout plan

Building 1 · Storey 1 · Room 6

Luminaire layout plan

Manufacturer	3F Filippi S.p.A.	P	21.0 W
Article No.	36576	Φ _{Luminaire}	2291 lm
Article name	Lucequadro LED 2000 EP VS		
Fitting	1x LED C - 2000 - 840		

15 x 3F Filippi S.p.A. Lucequadro LED 2000 EP VS

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.131 m / 0.392 m / 2.914 m	1.155 m	2.913 m	2.914 m	1
X-direction	5 pcs., Centre - centre, 2.251 m	3.405 m	2.892 m	2.914 m	2
		5.656 m	2.871 m	2.914 m	3
Y-direction	3 pcs., Centre - centre, 1.261 m	7.907 m	2.851 m	2.914 m	4
		10.158 m	2.830 m	2.914 m	5
Arrangement	A1	1.143 m	1.653 m	2.914 m	6
		3.394 m	1.632 m	2.914 m	7
		7.895 m	1.590 m	2.914 m	8
		10.146 m	1.569 m	2.914 m	9
		5.644 m	1.611 m	2.914 m	10
		1.131 m	0.392 m	2.914 m	11
		3.382 m	0.371 m	2.914 m	12
		5.633 m	0.350 m	2.914 m	13

Building 1 · Storey 1 · Room 6

Luminaire layout plan

X	Y	Mounting height	Luminaire
7.884 m	0.330 m	2.914 m	14
10.134 m	0.309 m	2.914 m	15

Building 1 · Storey 1 · Room 6

Luminaire list Φ_{total}

34365 lm

 P_{total}

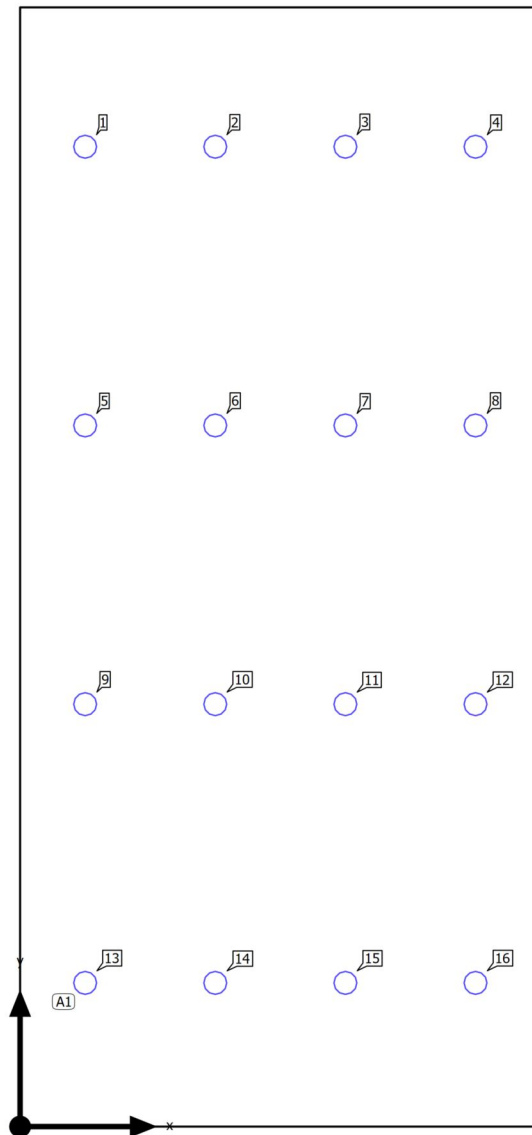
315.0 W

Luminous efficacy

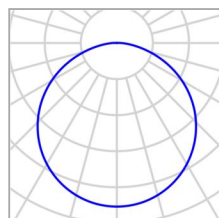
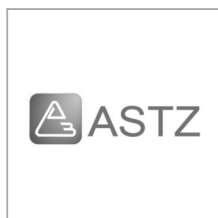
109.1 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
15	3F Filippi S.p.A.	36576	Lucequadro LED 2000 EP VS	21.0 W	2291 lm	109.1 lm/W

Building 1 · Storey 1 · Room 7

Luminaire layout plan

Building 1 · Storey 1 · Room 7

Luminaire layout plan

Manufacturer	ASTZ	P	22.5 W
Article No.	-	Φ _{Luminaire}	2701 lm
Article name	DBO85-24-241 Tablette EM1 840		
Fitting	1x LED		

16 x ASTZ DBO85-24-241 Tablette EM1 840

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.750 m / 1.656 m / 2.800 m	0.750 m	11.294 m	2.800 m	1
X-direction	4 pcs., Centre - centre, 1.500 m	2.250 m	11.294 m	2.800 m	2
		3.750 m	11.294 m	2.800 m	3
Y-direction	4 pcs., Centre - centre, 3.213 m	5.250 m	11.294 m	2.800 m	4
Arrangement	A1	0.750 m	8.081 m	2.800 m	5
		2.250 m	8.081 m	2.800 m	6
		3.750 m	8.081 m	2.800 m	7
		5.250 m	8.081 m	2.800 m	8
		0.750 m	4.869 m	2.800 m	9
		2.250 m	4.869 m	2.800 m	10
		3.750 m	4.869 m	2.800 m	11
		5.250 m	4.869 m	2.800 m	12
		0.750 m	1.656 m	2.800 m	13

Building 1 · Storey 1 · Room 7

Luminaire layout plan

X	Y	Mounting height	Luminaire
2.250 m	1.656 m	2.800 m	14
3.750 m	1.656 m	2.800 m	15
5.250 m	1.656 m	2.800 m	16

Building 1 · Storey 1 · Room 7

Luminaire list Φ_{total}

43216 lm

 P_{total}

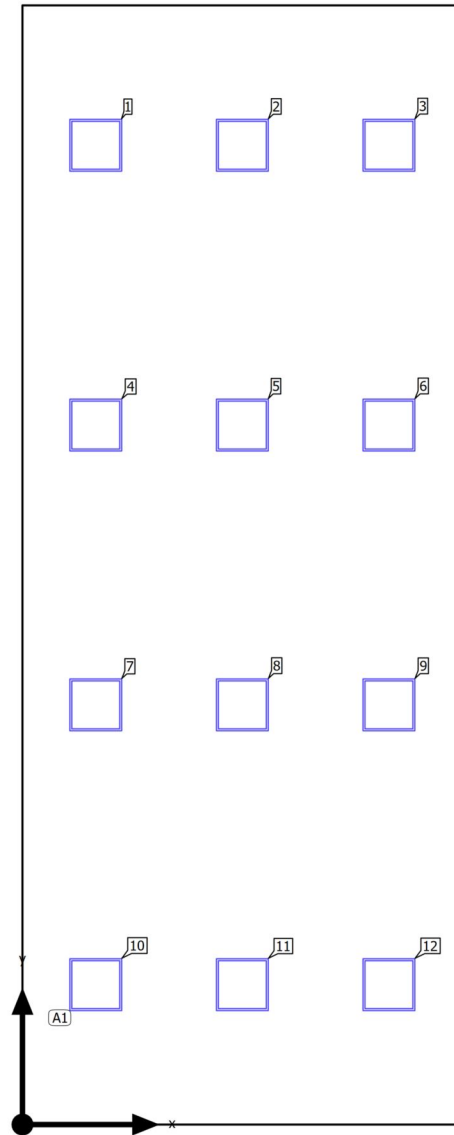
360.0 W

Luminous efficacy

120.0 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
16	ASTZ	-	DBO85-24-241 Tablette EM1 840	22.5 W	2701 lm	120.2 lm/W

Building 1 · Storey 1 · Room 8

Luminaire layout plan

Building 1 · Storey 1 · Room 8

Luminaire layout plan

Manufacturer	3F Filippi S.p.A.	P	34.0 W
Article No.	22790+A01485	Φ _{Luminaire}	3830 lm
Article name	3FLP6060UGR-830 (850mA) + ON-OFF		
Fitting	1x LED L - LED Panel - 2 - 830		

12 x 3F Filippi S.p.A. 3FLP6060UGR-830 (850mA) + ON-OFF

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.842 m / 1.606 m / 2.807 m	0.842 m	11.244 m	2.807 m	1
X-direction	3 pcs., Centre - centre, 1.683 m	2.525 m	11.244 m	2.807 m	2
		4.208 m	11.244 m	2.807 m	3
Y-direction	4 pcs., Centre - centre, 3.213 m	0.842 m	8.031 m	2.807 m	4
		2.525 m	8.031 m	2.807 m	5
		4.208 m	8.031 m	2.807 m	6
		0.842 m	4.819 m	2.807 m	7
		2.525 m	4.819 m	2.807 m	8
		4.208 m	4.819 m	2.807 m	9
		0.842 m	1.606 m	2.807 m	10
		2.525 m	1.606 m	2.807 m	11
		4.208 m	1.606 m	2.807 m	12

Building 1 · Storey 1 · Room 8

Luminaire list Φ_{total}

45960 lm

 P_{total}

408.0 W

Luminous efficacy

112.6 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
12	3F Filippi S.p.A.	22790+A0 1485	3FLP6060UGR-830 (850mA) + ON-OFF	34.0 W	3830 lm	112.6 lm/W

Glossary

A

A	Formula symbol for a surface in the geometry
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B

Background area	The background area borders the direct ambient area according to DIN EN 12464-1 and reaches up to the borders of the room. In larger rooms, the background area is at least 3 m wide. It is located horizontally at floor level.
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C

CCT	(Engl. correlated color temperature) Body temperature of a thermal radiator which serves to describe its light color. Unit: Kelvin [K]. The lesser the numerical value the redder; the greater the numerical value the bluer the light colour. The color temperature of gas-discharge lamps and semi-conductors are termed "correlated color temperature" in contrast to the color temperature of thermal radiators. Allocation of the light colors to the color temperature ranges acc. to EN 12464-1: Light color - color temperature [K] warm white (ww) < 3,300 K neutral white (nw) ≥ 3,300 – 5,300 K daylight white (dw) > 5,300 K
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Clearance height	The designation for the distance between upper edge of the floor and bottom edge of the ceiling (in the completely furnished status of room).
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Control group	A group of luminaires that are dimmed and controlled together. For each lighting scene, a control group provides its own dimming value. All luminaires within a control group share this dimming value. The control groups with their luminaires are automatically determined by DIALux on the basis of the created light scenes and their luminaire groups.
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CRI	(Engl. color rendering index) Designation for the color rendering index of a luminaire or a lamp acc. to DIN 6169: 1976 or CIE 13.3: 1995. The general color rendering index Ra (or CRI) is a dimensionless figure that describes the quality of a white light source in regards to its similarity with the remission spectra of defined 8 test colors (see DIN 6169 or CIE 1974) to a reference light source.
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Glossary

D

Daylight autonomy	Describes what percentage of the daily working time the required illuminance is met by daylight. The nominal illuminance is used from the room profile, unlike described in EN 17037. The calculation is not done in the centre of the room but at the placed sensor measuring point. A room is considered sufficiently supplied with daylight if it achieves at least 50% daylight autonomy.
Daylight factor	Ratio of the illuminance achieved solely by daylight incidence at a point in the inside to the horizontal illuminance in the outer area under an unobstructed sky. Formula symbol: D (Engl. daylight factor) Unit: %
Daylight quotient effective area	A calculation surface within which the daylight quotient is calculated.

E

Energy evaluation	Based on an hourly calculation procedure for daylight in indoor spaces, considering the project geometry and any existing daylight control systems. Orientation and location of the project are also considered. The calculation uses the specified system power of the luminaires to determine the energy demand. A linear relationship between power and luminous flux in the dimmed state is assumed for daylight-controlled luminaires. Times of use and nominal illuminance are determined from the usage profiles of the spaces. Switched-on luminaires that are explicitly excluded from control also consider the specified times-of-use. The daylight control systems use a simplified control logic that closes them at an outdoor horizontal illuminance of 27,500lx. The calendar year 2022 is used as a reference only. It is not a simulation of this year. The reference year is only used to assign the days of the week to the calculated results. The changeover to summer time is not considered. The reference sky type used is the average sky described in CIE 110 without direct sunlight. The method was developed together with the Fraunhofer Institute for Building Physics and is available for review by the Joint Working Group 1 ISO TC 274 as an extension of the previous annual regression-based method.
Environmental zones	The assessment of intrusive light and light immission depends on the environment of the lighting installation. Depending on the standard, 4-6 different zones are defined, ranging from highly protected areas in natural settings to urban areas, commercial zones, and industrial zones.
Eta (η)	(light output ratio) The light output ratio describes what percentage of the luminous flux of a free radiating lamp (or LED module) is emitted by the luminaire when installed. Unit: %

Glossary

G

g_1	Often also U_o (Engl. overall uniformity) Designates the overall uniformity of the illuminance on a surface. It is the quotient from E_{min} to $\bar{E}$ and is required, for instance, in standards for illumination of workstations.
g_2	Actually it designates the "non-uniformity" of the illuminance on a surface. It is the quotient of E_{min} to E_{max} and is generally only relevant for certifying the emergency lighting acc. to EN 1838.

I

Illuminance	Describes the ratio of the luminous flux that strikes a certain surface to the size of this surface ($lm/m^2 = lx$). The illuminance is not tied to an object surface. It can be determined anywhere in space (inside or outside). The illuminance is not a product feature because it is a recipient value. Luxometers are used for measuring. Unit: Lux Abbreviation: lx Formula symbol: E
Illuminance, adaptive	For the determining of the middle adaptive illuminance on a surface, this is rastered "adaptively". In the area of large illuminance differences within the surface, the raster is subdivided finer; within lesser differences, a rougher classification is made.
Illuminance, horizontal	Illuminance that is calculated or measured on a horizontal (level) surface (this can be for example a table top or the floor). The horizontal illuminance is usually identified by the formula letter E_h .
Illuminance, perpendicular	Illuminance that is calculated or measured plumb-vertical to a surface. This needs to be taken into account for tilted surfaces. If the surface is horizontal or vertical, then there is no difference between the perpendicular and the horizontal or vertical illuminance.
Illuminance, vertical	Illuminance that is calculated or measured on a vertical surface (this can be for example the front of some shelves). The vertical illuminance is usually identified by the formula letter E_v .

K

k_s	The glare effect of a light source can be described by the glare metric k_s . It relates the solid angle of the glaring light source as seen from the point of immission, the ambient luminance, and the maximum allowable luminance.
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Glossary

L

LENI	(Engl. lighting energy numeric indicator) Lighting energy numeric indicator acc. to EN 15193 Unit: kWh/(m² * a)
LLMF	(Engl. lamp lumen maintenance factor)/acc. to CIE 97: 2005 Lamp flux maintenance factor that takes the luminous flux reduction into account of a luminaire or an LED module in the course of the operating time. The lamp flux maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no luminous flux reduction existing).
LMF	(Engl. luminaire maintenance factor)/acc. to CIE 97: 2005 Luminaire maintenance factor that takes the soiling into account of the luminaire in the course of the operating time. The luminaire maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no soiling existing).
LSF	(Engl. lamp survival factor)/acc. to CIE 97: 2005 Lamp survival factor that takes the total failure into account of a luminaire in the course of the operating time. The lamp survival factor is specified as a decimal digit and can have a maximum value of 1 (no failures existing within the time concerned or prompt replacement after the failure).
Luminance	Dimension for the "brightness impression" that the human eye has of a surface. The surface itself can emit light thereby or light striking it can be reflected (emitter value). It is the only photometric value that the human eye can perceive. Unit: Candela per square metre Abbreviation: cd/m² Formula symbol: L
Luminous efficacy	Ratio of the emitted luminous flux Φ [lm] to the absorbed electrical power P [W] Unit: lm/W. This ratio can be formed for the lamp or LED module (lamp or module light output), the lamp or module with control gear (system light output) and the complete luminaire (luminaire light output).
Luminous flux	Dimension for the total light output that is emitted from one light source in all directions. It is thus an "emitter value" that specifies the entire emitting output. The luminous flux of a light source can only be determined in a laboratory. A difference is made between the lamp or LED module luminous flux and the luminaire luminous flux. Unit: Lumen Abbreviation: lm Formula symbol: Φ

Glossary

Luminous intensity	Describes the intensity of the light in a certain direction (emitter value). The luminous intensity is a matter of the luminous flux Φ that is emitted in a certain spherical angle Ω. The radiation characteristics of a light source are presented graphically in a light distribution curve (LDC). The luminous intensity is an SI base unit. Unit: Candela Abbreviation: cd Formula symbol: I
M	
Maintenance factor	See MF
MF	(Engl. maintenance factor)/acc. to CIE 97: 2005 Maintenance factor as decimal number between 0 and 1 that describes the ratio of the new value of a photometric planning parameter (e.g. of the illuminance) to a maintenance value after a certain time. The maintenance factor takes into account the soiling of luminaires and rooms as well as the luminous flux reduction and the failure of light sources. The maintenance factor is taken into account either overall or determined in detail acc. to CIE 97: 2005 by the formula $RMF \times LMF \times LLMF \times LSF$.
O	
Obtrusive light/Light immission	To protect the nocturnal environment and minimize problems for humans, flora, and fauna, it is necessary to limit obtrusive light (also known as light pollution), which can cause serious physiological and ecological issues for individuals and the environment. Light immission refers to the disturbing influence of emitted light from artificial light sources.
Operating times	The assessment of obtrusive light and light immission depends on the operating times of the lighting installation. Depending on the standard, 1-3 different operating times are specified. In the absence of specific details, an operating time between 06:00 and 22:00 can be assumed.
P	
P	(Engl. power) Electric power consumption Unit: watt Abbreviation: W

Glossary

R

$R_{(UG)} \text{ max}$	Measure of the psychological glare in indoor spaces. In addition to the luminance of luminaires, the level of the $R_{(UG)}$ value also depends on the observer position, the viewing direction and the ambient luminance. The calculation is made according to the table method, see CIE 117. Among other things, EN 12464-1:2021 specifies maximum permissible $R_{(UG)}$- values $R_{(UGL)}$ for various indoor workplaces.
R_{DLO}	The ratio of the luminous flux emitted below the horizontal plane to the total lamp luminous flux of a luminaire or lighting installation in its operational position.
R_G	The glare directly caused by luminaires of an outdoor lighting installation is determined using the CIE Glare Rating (RG) method. To calculate this, the equivalent veiling luminance of the surroundings is needed. There are four options for determining this: • An exact calculation according to CIE 112, based on the scene area. • A simplified method according to EN 12464-2, based on the scene area. • Using a custom calculation area to determine the equivalent veiling luminance. • Specifying a fixed value for easy comparability.
R_{UF}	upward flux ratio The ratio of the luminous flux emitted directly or reflected above the horizontal plane to the luminous flux that cannot be avoided under ideal conditions to achieve the illuminance level on a deliberately illuminated area.
R_{UL}	upward light ratio The ratio of the luminous flux emitted above the horizontal plane to the luminous flux of a luminaire or lighting installation in its operational position. The luminaire efficiency is considered in this calculation.
R_{ULO}	upward light output ratio The ratio of the luminous flux emitted above the horizontal plane to the total lamp luminous flux of a luminaire or lighting installation in its operational position.
Reflection factor	The reflection factor of a surface describes how much of the striking light is reflected back. The reflection factor is defined by the color of the surface.
RMF	(Engl. room maintenance factor)/acc. to CIE 97: 2005 Room maintenance factor that takes the soiling into account of the space encompassing surfaces in the course of the operating time. The room maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no soiling existing).
$RUG \text{ (max)}$	(unified glare rating) Measure for the psychological glare effect in interiors. In addition to luminaire luminance, the RUG value also depends on the position of the observer, the viewing direction and the ambient luminance. Among other things, EN 12464-1 specifies maximum permissible RUG values for various indoor workplaces.

Glossary

RUG observer	Calculation point in the room, for the DIALux the RUG value is determined. The location and height of the calculation point should correspond to the typical observer position (position and eye level of the user).
S	
Surrounding area	The ambient area directly borders the area of the visual task and should be planned with a width of at least 0.5 m according to DIN EN 12464-1. It is at the same height as the area of the visual task.
V	
Visual task area	The area that is needed for carrying out the visual task in accordance with DIN EN 12464-1. The height corresponds with the height at which the visual task is executed.
W	
Wall zone	Circumferential area between working plane and walls which is not taken into account for the calculation.
Working plane	Virtual measuring or calculation surface at the height of the visual task that generally follows the room geometry. The working plane may also feature a wall zone.